ChatGPT

ICECAP-A: Capability Measure for Adults – Domains, Applications, and Validity

Overview of ICECAP-A

The **ICEpop CAPability measure for Adults (ICECAP-A)** is a generic **well-being instrument** grounded in the capability approach. Developed in 2012, it shifts focus from traditional health outcomes to a broader view of quality of life by quantifying what people are able to do and be in key life domains [1] [2]. Unlike symptom-based measures, ICECAP-A evaluates an individual's capacity to live a meaningful life beyond just being disease-free [2]. It consists of a short self-report questionnaire (5 items, each with four response levels) that covers five fundamental **capability dimensions** important for adult well-being. These dimensions together produce an index value (0–1 scale) representing overall capability, which can be used in economic evaluations (e.g. calculating **"Years of Full Capability"** analogous to QALYs) [3] [4].

Five Capability Dimensions of ICECAP-A

ICECAP-A captures well-being through **five capability dimensions**, each reflecting an essential aspect of life quality. Below are the five domains and their meanings [5]:

- **Stability** – the ability to feel settled and secure in life [5]. This refers to having a sense of safety and stability in one's day-to-day circumstances (e.g. financial and personal security).
- **Attachment** – the ability to have love, friendship, and support [5]. This covers meaningful social connections and personal relationships that provide emotional support.
- **Autonomy** – the ability to be independent and make one's own decisions [5]. It reflects having control over one's life and freedom to choose how to live.
- **Achievement** – the ability to achieve and progress in life [5]. This entails a sense of accomplishment or personal growth, such as pursuing goals and feeling that one's activities are worthwhile.
- **Enjoyment** – the ability to experience enjoyment and pleasure [5]. This dimension captures everyday happiness, contentment, and the capacity to enjoy life's experiences.

Each ICECAP-A item asks the respondent to indicate their level of capability in that domain (from full capability to no capability). Higher levels on all five domains imply a greater overall capability well-being [6].

Applications in Mental Health and Employment Support

ICECAP-A's broad focus on life capabilities makes it particularly useful in **mental health and social care settings**, where interventions often aim to improve overall well-being rather than just physical health. Indeed, ICECAP-A has been **applied in mental health research** and shown value in capturing outcomes for people with mental illness:

- **Depression:** A study in the UK assessed ICECAP-A in adults with depression and found it valid for this population [7]. The ICECAP-A scores were meaningfully associated with mental health status – for example, worse depression and psychological distress corresponded to lower ICECAP-A

capability, as hypothesized [7]. Notably, depression-specific scales (e.g. DASS-21 depression and K10 distress) explained more of the variation in ICECAP-A scores than a generic health utility (EQ-5D) did [7]. This suggests ICECAP-A is sensitive to aspects of well-being impacted by mental health that traditional health QoL measures might miss. The authors concluded that ICECAP-A is **suitable for outcome assessment in adults with depression**, and using a capability perspective could better inform resource allocation in mental health care [8] [9].

- **Substance Use/Addiction:** ICECAP-A has also been tested in the context of addiction recovery. In a study of individuals with opioid dependence, ICECAP-A demonstrated strong **construct validity and responsiveness** [10]. It correlated with clinical indicators of well-being and picked up changes over time in both health status and psychosocial functioning [11]. In fact, ICECAP-A was **more sensitive to changes** in recovery outcomes than the EQ-5D-5L, which showed ceiling effects in this group [11]. These findings indicate that ICECAP-A can capture the broader improvements in life situation (e.g. social stability, sense of purpose) that mark recovery, beyond just health symptom changes. The researchers noted that ICECAP-A "gives more attention to broader impacts" and shows promise for evaluating substance use disorder treatments where achieving a better life (not merely symptom reduction) is the goal [12].

- **Other Mental Health Contexts:** Generally, ICECAP-A and its elder variant ICECAP-O have been used in various mental illness studies (e.g. serious depression, substance abuse, etc.) [13]. A systematic review (2020) confirmed growing feasibility of ICECAP-A in such contexts and called for further psychometric research [14]. The development of **mental health-specific capability measures** like OxCAP-MH (Oxford CAPability for Mental Health) also underlines the interest in capability-based assessment for psychological well-being [15]. ICECAP-A itself, while not disorder-specific, provides a **holistic outcome measure** for mental health interventions – focusing on whether treatments help individuals attain a life with supportive relationships, autonomy, and fulfillment, rather than only symptom relief [16]. For example, psychosocial programs (therapy, community support, etc.) often aim to improve a person's sense of control, social connection, or life satisfaction – outcomes that ICECAP-A is designed to value [16].

- **Employment Support:** In the domain of **vocational rehabilitation and employment support** (支援就労領域), the use of ICECAP-A is conceptually very relevant, though documented cases are still emerging. Programs that help individuals with disabilities or mental health conditions find and keep employment have impacts beyond income – they can enhance independence, social inclusion, and self-worth (paralleling improvements in ICECAP-A dimensions like Autonomy, Attachment, Achievement, and Stability). **Policy experts have suggested using capability measures to evaluate such social interventions**: for instance, health technology assessment bodies in the UK and Netherlands recommend including capability-based outcomes when an intervention's benefits are largely social or non-health [17]. By capturing changes in a person's life opportunities and well-being, ICECAP-A could demonstrate the full value of employment support initiatives that might not register on health-only scales. However, our research did not find specific published studies in which ICECAP-A was applied to an employment support program – indicating a **potential area for future research**. It is reasonable to expect that as ICECAP-A gains traction, it will be used alongside or in place of traditional quality-of-life measures to assess outcomes of vocational training, supported employment, and related services. In Japan, where the social integration of individuals with disabilities is a pressing issue, **introducing ICECAP-A in employment support evaluation** could help quantify improvements in clients' life capabilities (e.g. feeling more secure, having a sense of achievement) beyond the economic benefits.

(In summary, ICECAP-A is increasingly utilized in mental health and social care research to capture outcomes that matter to service users' lives. Its emphasis on empowerment, opportunity, and overall well-being makes it a valuable tool in areas like mental health support, social care, and potentially employment assistance, where traditional clinical metrics often overlook these broader benefits [16].)

Comparison with QoL Measures (EQ-5D, etc.)

How ICECAP-A differs: Traditional **health-related quality of life (HRQoL)** instruments (e.g. the EQ-5D or SF-36) focus on health status and functional impairments. In contrast, ICECAP-A is a **broader measure of quality of life** grounded in the capability approach, not tied strictly to health or disability [1]. It treats good health as one factor influencing well-being, but explicitly includes non-health domains (like enjoyment or social connection) which standard HRQoL tools might neglect [1] [18]. In essence, HRQoL tools assess what people actually do or feel (given their health), whereas ICECAP-A assesses what people are able to do or be, considering all aspects of life. This broader scope means ICECAP-A can complement health indices, especially in contexts (such as **mental health, social care, or aging**) where important outcomes (happiness, dignity, relationships) are not fully captured by health status questions.

Empirical comparisons: Studies comparing ICECAP-A with conventional QoL measures find that they provide **related but distinct information**. For example, a psychometric study in Chinese patients with type 2 diabetes found only a **weak correlation** between ICECAP-A index scores and EQ-5D utilities (Spearman $r \approx 0.12$) [19]. An exploratory factor analysis showed that ICECAP-A's five attributes loaded onto a different factor than the EQ-5D's health dimensions, indicating that capability well-being represents a separate construct from health-related quality of life [20]. This suggests ICECAP-A is capturing aspects of quality of life that EQ-5D does not. Indeed, the authors concluded that **ICECAP-A measures additional dimensions of QoL beyond traditional HRQoL instruments** [21]. They recommend using both types of measures together to get a more complete picture of patient well-being [22]. Similarly, in the depression study mentioned earlier, the ICECAP-A was more strongly associated with mental health measures than the EQ-5D-5L was, underscoring that **capability well-being and health status are related but not identical** aspects of quality of life [7].

When to use which: There is ongoing discussion on how to choose between or combine ICECAP and other QoL scales. One approach is to use **ICECAP-A alongside a health-focused QoL measure** in evaluations. This was exemplified in a UK trial for addiction: ICECAP-A detected improvements in well-being (e.g. social role, satisfaction) that EQ-5D missed due to ceiling effects, whereas EQ-5D uniquely captured some health symptoms – together providing a holistic assessment [10]. In contexts like **long-term care or mental health**, where interventions aim to improve life satisfaction, social connection, or functioning, ICECAP-A may better reflect those benefits than a narrow health utility score [23]. However, ICECAP-A is not a direct replacement for health QoL measures when the interest is specifically clinical outcomes; rather, it **expands the evaluative space** to include well-being outcomes. This has led some to propose new evaluation frameworks (e.g. calculating capability-adjusted life years in parallel to QALYs) for economic appraisal of interventions that affect broader quality of life [24] [25].

In summary, **QoL vs Capability** measures serve different purposes: HRQoL instruments (like EQ-5D, SF-36) excel at capturing health status and functional limitations, which is crucial for many medical evaluations. ICECAP-A captures the value of being well – feeling secure, happy, autonomous, supported – which may resonate more with patient-centered outcomes and social care benefits. Using them jointly or selecting the appropriate one depends on the context. For instance, in a cost-utility analysis of a purely medical intervention, EQ-5D might suffice; but for a psychosocial intervention or public health program where outcomes include social well-being or empowerment, ICECAP-A would provide critical insight [16] [25].

ICECAP-A and WHODAS (Disability Assessment)

WHODAS 2.0 (WHO Disability Assessment Schedule) is a widely used instrument to measure disability and functional impairment across domains like cognition, mobility, self-care, interpersonal relations, life activities, and participation. It is based on the WHO's International Classification of Functioning (ICF) framework. Comparatively, ICECAP-A and WHODAS represent two different perspectives: - **WHODAS** evaluates the **degree of difficulty** an individual has in performing daily activities and societal participation (essentially, it quantifies functional limitations or disability). It asks, for example, how much one's health problems affect doing household chores, joining community activities, concentrating, dressing, etc. The results reflect a person's actual functioning and restrictions in the past 30 days, providing a disability score. - **ICECAP-A** evaluates one's **capability to achieve well-being** in key life domains, regardless of health condition. It focuses on whether the person can lead a fulfilling life in areas of security, autonomy, relationships, achievement, and enjoyment – essentially a self-evaluation of life opportunities and quality, rather than the obstacles.

Because of these differences, **ICECAP-A and WHODAS are not directly interchangeable** but rather complementary. WHODAS is very useful for clinical and rehabilitation settings to gauge how much a health condition (or disability) is impeding a person's functioning in daily life. ICECAP-A, on the other hand, is useful for capturing the person's overall well-being or quality of life **achieved or achievable**, which might be high even if some disabilities are present (for example, someone with a disability might report good capability well-being if they have strong social support, autonomy through assistive devices, etc.).

In practice, using both tools can provide a **holistic picture**: WHODAS can identify needs for support by pinpointing areas of severe functional limitation, while ICECAP-A can reflect whether interventions (medical, social, or environmental) are enabling the individual to live a life they value. In mental health contexts, for instance, WHODAS might quantify impairment in work or social activities due to illness, while ICECAP-A might tell us whether the person still feels a sense of enjoyment or attachment despite those challenges. If a treatment greatly reduces WHODAS disability scores but ICECAP-A remains low, it suggests unmet needs in the person's well-being (perhaps loneliness or lack of purpose). Conversely, high ICECAP-A alongside persistent WHODAS limitations might indicate successful adaptation or support systems that allow the person to flourish despite functional impairments.

There is limited direct research comparing ICECAP-A and WHODAS head-to-head. However, conceptually, **WHODAS aligns with a "functionings" approach (what one actually does/can do in daily activities), while ICECAP aligns with the "capabilities" approach (the freedom to do/be certain things)** [26]**. Both are important in disability and mental health research. Scholars have even suggested that combining the ICF framework (which underpins WHODAS) with the capability approach could foster more patient-centered, holistic care** [27]**. For practitioners, the choice may come down to focus: use** WHODAS **when the goal is to assess and address functional impairments and need for support in specific domains of life; use** ICECAP-A** when the goal is to assess overall life quality, values, and well-being outcomes of interventions. In some cases, employing both will yield the richest understanding of intervention impacts – one instrument showing improvements in functional ability and the other capturing gains in life satisfaction and capability.

(In summary, WHODAS measures "how much difficulty" a person has in life domains due to health conditions, whereas ICECAP-A measures "how good a life" a person is able to lead across domains. Both contribute to understanding quality of life: WHODAS from a disability perspective, and ICECAP-A from a well-being and opportunity perspective.)

Structural Validity and Cultural Adaptation of ICECAP-A

Structural validity: Since its development, ICECAP-A has undergone multiple psychometric evaluations to test whether it truly measures a cohesive construct of "capability well-being" and how its items perform. Research generally supports ICECAP-A's **construct validity** – i.e., that it captures a meaningful latent trait related to quality of life. For instance, internal consistency of the five items tends to be acceptable (Cronbach's α around 0.7 in population samples) [19], suggesting the domains hang together moderately well. Some studies employ **factor analysis** to examine ICECAP-A's dimensionality. As noted, ICECAP-A is often modeled as a single index (one overarching factor of capability). Empirical data from different countries back this: e.g., in a Chinese sample, all ICECAP-A items were associated with one factor separate from health items [20], and in a Hungarian general population, confirmatory factor analysis also supported a unidimensional structure for ICECAP-A [28]. That said, minor issues have been observed – a study in dermatology patients found a local dependence between the **Attachment** and **Enjoyment** items (meaning those two were closely linked beyond the general factor) [29]. This could be due to these domains both relating to emotional well-being. Overall, ICECAP-A scores show expected correlations with other measures: **strong correlations with life satisfaction and happiness**, but weaker correlations with health-specific QoL or symptom indices, indicating it taps into broader well-being rather than health status [30] [20]. ICECAP-A has also demonstrated **known-groups validity**: it can distinguish between groups with differing social and health circumstances. For example, people with higher income, employment, or better self-rated health tend to report higher ICECAP-A capabilities, as shown in several studies [31]. This aligns with theoretical expectations (since poverty, unemployment, and poor health constrain one's capabilities). Additionally, early evidence on **responsiveness** (sensitivity to change) is promising – ICECAP-A detected changes in capability after interventions in some trials (e.g. addiction treatment) more readily than a health utility did [10], though more research is needed on responsiveness in various contexts [32].

Cultural adaptation and validity across cultures: One important aspect of ICECAP-A's validity is whether its five domains are relevant and interpreted consistently in different cultural contexts. The instrument has been translated into numerous languages (including **Japanese, Chinese, Dutch, German, French, Persian, Hungarian, among others**) [33]. Each translation typically undergoes careful adaptation and validation. Researchers have generally found that the concepts of stability, attachment, autonomy, achievement, and enjoyment are meaningful across diverse cultures, but the **relative importance of these domains can differ by culture**. For example, **Japan** recently developed its own version of ICECAP-A and a corresponding Japanese value set. In the Japanese valuation study, all five attributes were confirmed to contribute to overall capability well-being [34]. However, **Japanese respondents valued certain capabilities differently** than European respondents. Notably, "Enjoyment" was ranked as the most important domain in Japan, whereas in UK studies "Attachment" (love and friendship) was ranked most important [35]. Japanese participants placed "Attachment" third, after Stability [35]. One hypothesis is that in Japanese culture – which emphasizes collective harmony and unspoken understanding in relationships – love and friendship might be taken for granted as a baseline, rather than explicitly rated as top priorities [36]. In other words, social support is viewed as implicit in group life, potentially leading to a lower explicit ranking of the attachment dimension in surveys [36]. This contrasts with Western contexts where individuals may more consciously prioritize personal relationships when considering quality of life. Such cultural nuances underscore why **developing country-specific tariffs and translations** is crucial: preferences can vary, and a direct import of a UK value set to Japan (or elsewhere) might misestimate well-being if, say, enjoyment is intrinsically valued more by one population and attachment by another [37]. The Japanese study also found differences in how people view improvements: Japanese respondents placed great weight on avoiding the worst levels of capability (level 1) – they gave a big jump in value from level 1 to 2 – but were less differentiating at the top end (levels 3 to 4) [38]. This suggests a tendency to **emphasize eliminating severe suffering over**

maximizing excellence, a pattern that was more pronounced than in the UK data [39]. It highlights how cultural attitudes (possibly influenced by collectivist ethos or loss-aversion) can affect responses on a capability measure.

Other national adaptations have also been pursued. A **Persian (Iranian) version** of ICECAP-A was cross-culturally adapted and tested, showing good psychometric properties in the Iranian general population (with some adjustments for linguistic equivalence) [40]. A **Chinese version** for mainland China required slight wording modifications to fit cultural context, but ultimately was found valid and reliable for Chinese patients [41] [19]. A Hungarian version was validated with population norms established [42], and a Dutch tariff was created to reflect Dutch general population preferences [43]. Moreover, **test–retest reliability** has been confirmed in at least one culture: a Danish study reported stable ICECAP-A scores over a short interval, indicating the instrument produces consistent results when no change in capability is expected [44]. All these efforts indicate a broad interest in making ICECAP-A usable globally, and generally they find that while minor cultural adaptations in phrasing or valuation are needed, the **underlying construct of capability well-being is robust across cultures**.

For **Japan**, specifically, the availability of a validated ICECAP-A (as of 2025) opens the door for its use in Japanese research and policy. Sado et al. (2025) not only derived a Japanese scoring system but also highlighted that using ICECAP-A can facilitate **more precise evaluation of quality of life for populations such as those with chronic health conditions, caregivers, and the general public** in Japan [45]. This suggests Japanese health economists and clinicians are recognizing the value of capability measures in areas like elderly care, mental health (caregivers often implies mental health burden), and community health where conventional QoL metrics might be insufficient. Ongoing research is encouraged to test ICECAP-A's **sensitivity over time in Japan** (e.g., detecting QoL changes with treatment) [46], and to continue exploring how cultural context influences responses. Given the initial findings, Japan's example reinforces that **one size may not fit all in quality-of-life measurement** – but with culturally attuned adaptation, ICECAP-A can reliably be used to compare and evaluate well-being outcomes within each society's own value framework.

Conclusion

ICECAP-A provides a **capability-based lens** to assess quality of life, broadening the evaluation beyond health to include fundamental aspects of well-being. It defines quality of life in terms of what people **can be and do** – feeling secure, loved, autonomous, accomplished, and joyful. This approach has proven especially relevant in fields like mental health, social care, and other interventions where improvements in life context and satisfaction are the primary goals rather than purely medical outcomes. Studies in mental health (depression, addiction) show ICECAP-A captures meaningful changes that traditional QoL scales might overlook, and its growing adoption internationally (including a Japanese version) attests to its adaptability and importance.

When comparing ICECAP-A with standard QoL instruments and disability scales, it's clear that each tool has its role: ICECAP-A **complements** health-focused measures by capturing the broader quality of life, whereas instruments like WHODAS quantify functional impairments. The choice of measure should align with the intended outcomes of interest – whether they be clinical health status, functional ability, or holistic well-being. Increasingly, practitioners and policymakers are considering **combined approaches** to truly understand the impact of health and social programs on people's lives [22] [25]. The capability approach embodied by ICECAP-A challenges us to value aspects of life that matter most to individuals, thus encouraging more person-centered and comprehensive decision-making in healthcare and welfare [47].

In summary, **ICECAP-A** offers a structured yet flexible way to quantify adult quality of life in a broad sense. Its five domains provide a succinct checklist of core life capabilities, and its usage is backed by a growing body of validation evidence. As research continues (especially on responsiveness and long-term changes), ICECAP-A is poised to become a standard outcome measure not only in cost-utility analysis but also in program evaluations for mental health services, elder care, disability support, and beyond – anywhere that **"quality of life"** is more than just a health statistic, but a reflection of one's freedom and opportunity to live well.

Sources:

- Rencz, F. et al. (2022). Measurement properties of the ICECAP-A capability well-being instrument among dermatological patients. Quality of Life Research, 31(3): 903-915 [48] [49].
- Mitchell, P.M. et al. (2017). Assessing the validity of the ICECAP-A capability measure for adults with depression. BMC Psychiatry, 17(1): 46 [7].
- Goranitis, I. et al. (2016). Measuring health and broader well-being benefits in the context of opiate dependence: ICECAP-A and EQ-5D-5L. Value in Health, 19(6): 820-828 [10].
- Xiong, Y. et al. (2021). Assessing the reliability and validity of the ICECAP-A in Chinese type 2 diabetes patients. Health and Quality of Life Outcomes, 19(1): 7 [19] [21].
- Sado, M. et al. (2025). Scoring system for the Japanese version of ICECAP-A. Psychology Research and Behavior Management, 18: 919-931 [50] [35].
- Marksman Healthcare (2025). The ICECAP-A instrument for capabilities – overview and insights [4] [16].

---

[1] [2] [4] [16] [18] [24] [25] [43] [47] The ICEpop Capability Measure for Adults (ICECAP-A) Instrument for Capabilities - Marksman Healthcare
https://marksmanhealthcare.com/2025/05/20/the-icepop-capability-measure-for-adults-icecap-a-instrument-for-capabilities/

[3] [5] [6] [13] [14] [15] [17] [26] [28] [29] [30] [31] [48] [49] Measurement properties of the ICECAP-A capability well-being instrument among dermatological patients - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC8921030/

[7] [8] [9] [32] Assessing the validity of the ICECAP-A capability measure for adults with depression - PubMed
https://pubmed.ncbi.nlm.nih.gov/28148234/

[10] [11] [12] [23] Measuring Health and Broader Well-Being Benefits in the Context of Opiate Dependence: The Psychometric Performance of the ICECAP-A and the EQ-5D-5L - PubMed
https://pubmed.ncbi.nlm.nih.gov/27712710/

[19] [20] [21] [22] Assessing the reliability and validity of the ICECAP-A instrument in Chinese type 2 diabetes patients - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC7788876/

[27] Toward a paradigm shift in healthcare: using the International ...
https://research.vu.nl/en/publications/toward-a-paradigm-shift-in-healthcare-using-the-international-cla/

[33] [40] [42] [44] ICECAP-A (ICEPop CAPability measure for Adults) | Mechanisms Underlying Mind-Body Interventions & Measurement of Emotional Well-Being: M3EWB
https://m3ewb.research.uconn.edu/measure/icecap-a-icepop-capability-measure-for-adults/

[34] [35] [36] [37] [38] [39] [45] [46] [50] Scoring System for the Japanese Version of the ICECAP-A - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC11932030/

41 Adaptation and assessments of the Chinese version of the ICECAP ...
https://link.springer.com/article/10.1186/s12955-018-0865-3